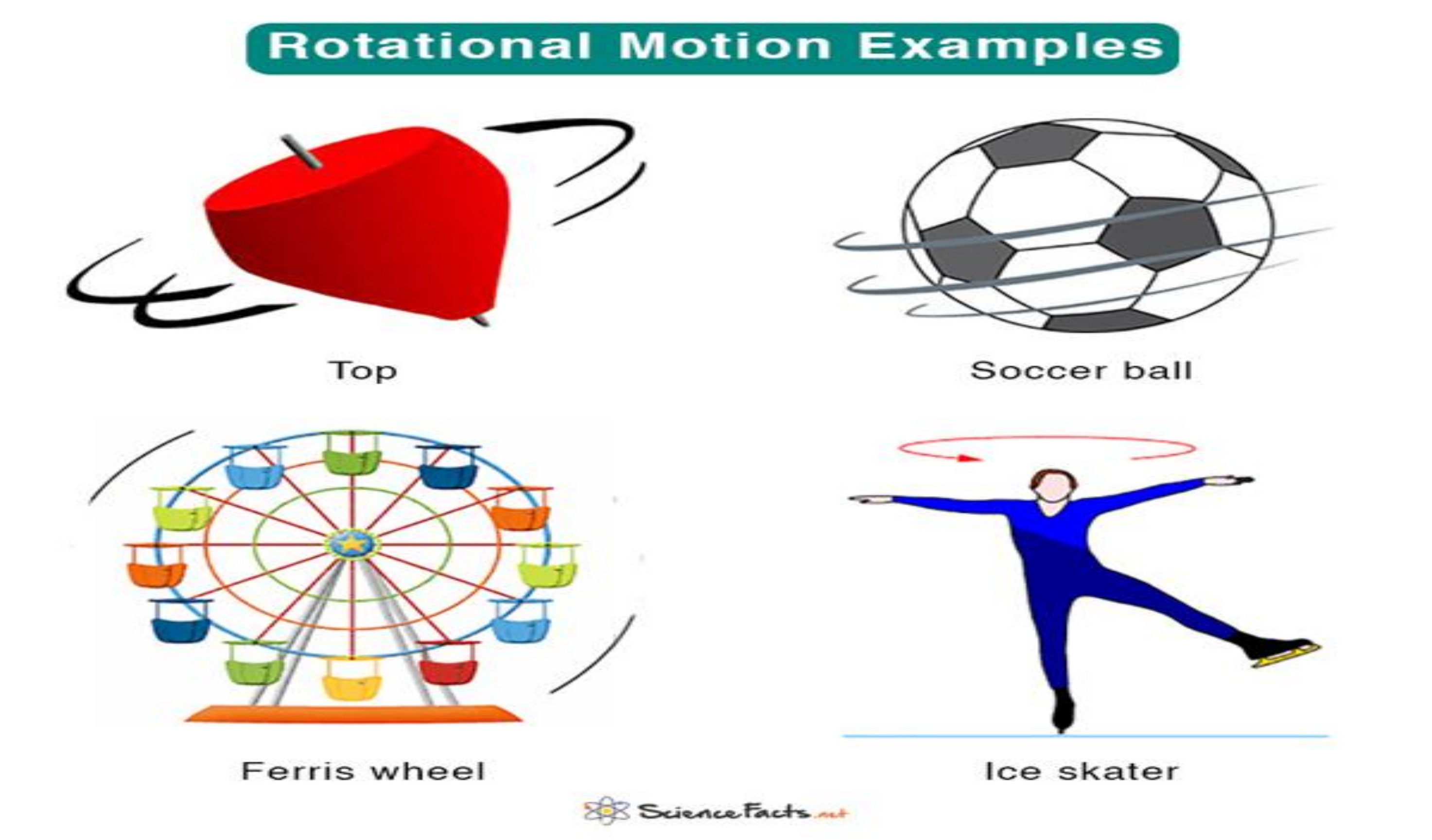
Experiment #(7)

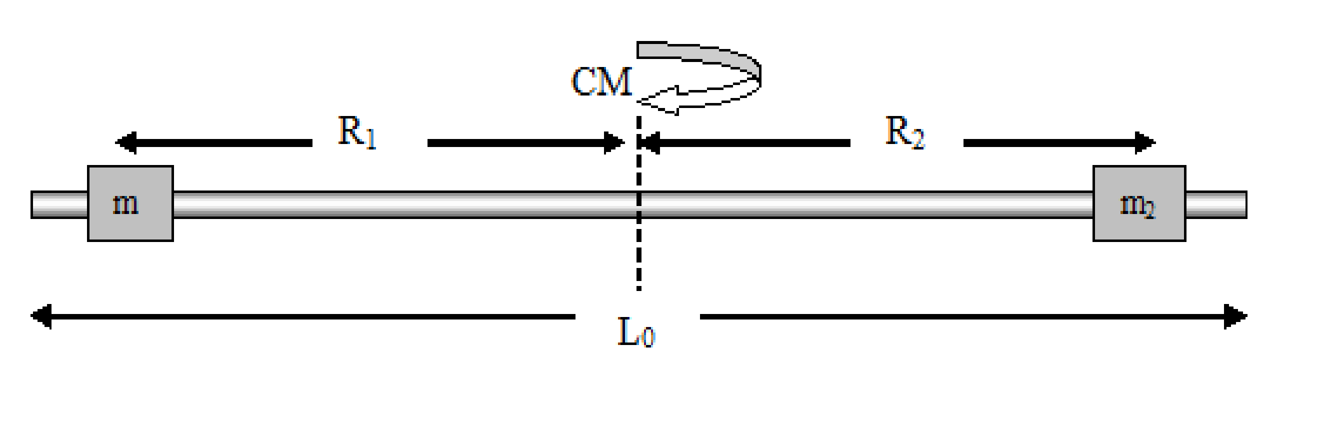
Rotational Dynamics

(ديناميكا الدوران)



In the following system the moment of inertia of the system is given by:

$$I = I_{CM} + m_1 R_1^2 + m_2 R_2^2$$



In special case if

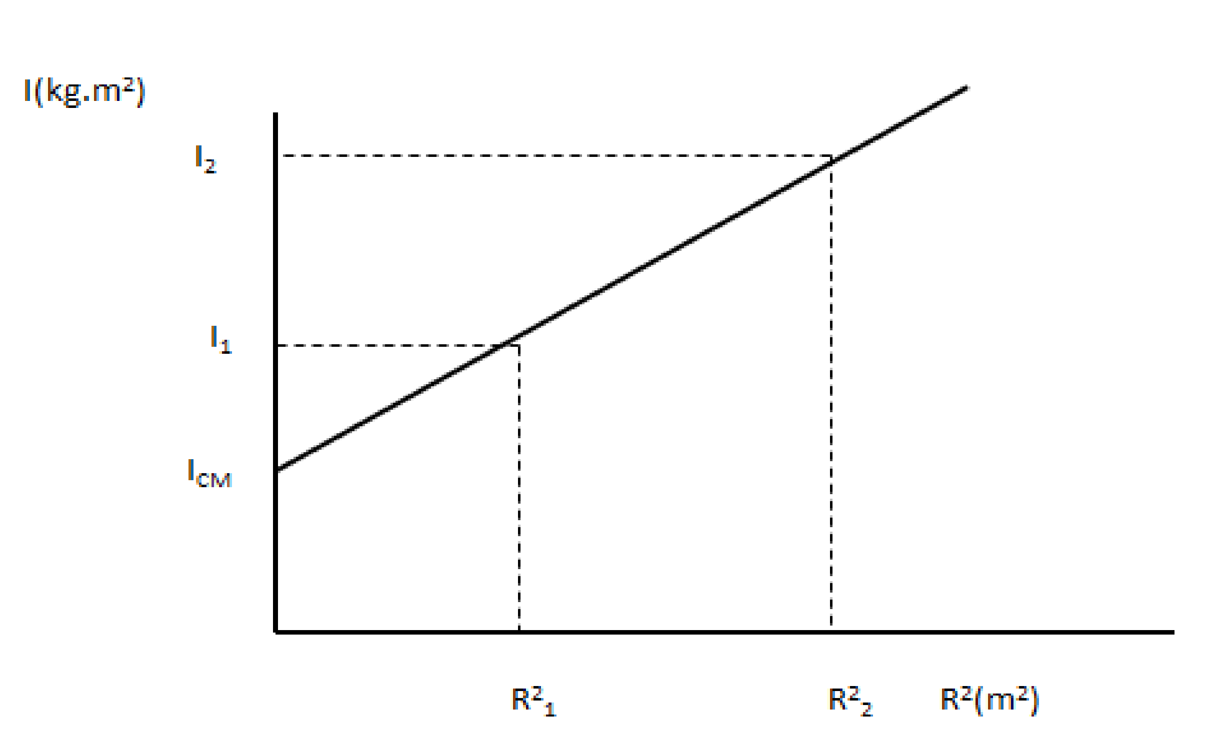
$m_1 = m_2 = m$ and $R_1 = R_2 = R$,

then

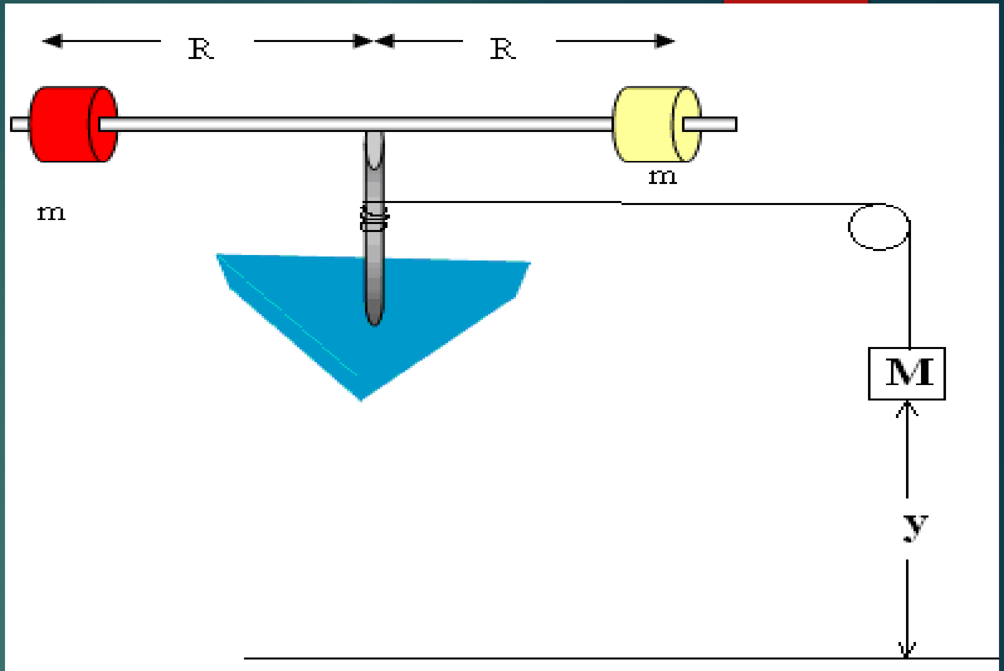
$$I = I_{CM} + 2mR^2$$

$$slope = 2m$$

$$m = \frac{slope}{2}$$



If the system is released from rest, then the mass $\boldsymbol{M}$ moves downward while the two masses, the rod connecting them and the cylinder of radius r start rotating. The rotational kinetic energy of the system is given by $\frac{1}{2}\boldsymbol{I_{tot}w^2}$ where $\boldsymbol{I_{tot}}$ is the total moment of inertia about the axis of rotation and $\boldsymbol{w}$ is the angular velocity. The total moment of inertia can be approximately written as $\boldsymbol{I_{tot} = I_{rod} + m_1R^2 + m_2R^2}$. From the work and kinetic energy theorem, we can show that $\boldsymbol{Mgy = \frac{1}{2}Mv^2 + \frac{1}{2}I_{tot}w^2 + n_1W_{fr}}$ where $\boldsymbol{n_1}$ is the number of revolutions of the rotating objects, $\boldsymbol{n_1W_{fr}}$ is the work done by friction forces and $\boldsymbol{v}$ is the velocity of $\boldsymbol{M}$ just **before it hits the ground**. After the mass $\boldsymbol{M}$ touches the ground, the system continues to rotate until the rotational kinetic energy is completely dissipated by friction forces. If we know the number of revolutions of the rotating object after $\boldsymbol{M}$ touches the ground $\boldsymbol{n_2}$, then $\boldsymbol{\frac{1}{2}I_{tot}w^2 = n_2W_{fr}}$. Using the equation $\boldsymbol{v = \frac{2y}{t}}$ and the relationship $\boldsymbol{v = rw}$, the above equations can be written as



$$\boldsymbol{I_{tot} = Mr^2\left[\frac{n_2}{n_1+n_2}\right]\left[\frac{gt^2}{2y} - 1\right]} \;\ldots\ldots\ldots\ldots(1)$$

Procedure :

- Measure the radius of the cylinder and the height of the mass **M** from the ground.
- Use two equal masses $m_1 = m_2 = m$
- Make sure that the length of the string around the cylinder equals the height **y,** so that the string leaves the cylinder when the mass M reaches the ground.
- Release the system from rest and measure n_1 , n_2 and t (the time required for the mass M to touch the ground).
- Repeat the experiment for five different values of **R**.
- Calculate the total moment of inertia I_{tot} from **equation 1**.
- tabulate your results in the table shown below.
- Plot I_{tot} versus R^2 on a graph paper and draw the best straight line through the points.
- Find the slope. Using the equation $I_{tot} = I_{rod} + m_1R^2 + m_2R^2$ and the calculated value of the slope, calculate $m = m_1 = m_2$.
- Weight the mass m and compare it with the calculated value in the last step.
- Calculate the error in m.
- From the curve, calculate the value of I_{rod}.
- Weight the rod and measure its length to calculate I_{rod} using $I_{rod} = \frac{1}{12} m_{rod}\, l^2$ and compare it with the calculated value in the last step.
- Calculate the error in I_{rod}.
- What are the possible sources of errors in this experiment?

$r =$ ( )					$y =$ ( )					
Trial	R ( )	R^2 ( )	n_1	n_2	t_1 ( )	t_2 ( )	t_3 ( )	t_{avg} ( )	t^2 ( )	I_{tot} ( )
1										
2										
3										
4										
5										

THANK YOU